

Duality

Lecture 4

October 1, 2026

Motivation

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4. *You are offered a bit more of b_i , for a “suitable price”. Is the deal worthwhile?*

Duality theory will provide answers to these questions (and more)

Outline

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- In the process, will uncover some **fundamental ideas in optimization**:

separation of convex sets $\implies$ Farkas Lemma $\implies$ strong duality

Deriving Lower Bounds

Consider a linear optimization problem in the most general form possible:

Primal Problem

$$\begin{array}{ll} (\mathcal{P}) \text{ minimize}_x & c^T x \\ \text{such that} & a_i^T x \geq b_i, \quad \forall i \in I_{ge}, \\ & a_i^T x \leq b_i, \quad \forall i \in I_{le}, \\ & a_i^T x = b_i, \quad \forall i \in I_{eq}, \\ & x_j \geq 0, \quad \forall j \in J_p, \\ & x_j \leq 0, \quad \forall j \in J_n, \\ & x_j \text{ free}, \quad \forall j \in J_f \\ \text{variable} & x \in \mathbb{R}^n. \end{array} \tag{1}$$

Note the mnemonic encoding...

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Definition

We will refer to this as the **primal problem** or problem $(\mathcal{P})$.

Let P denote its feasible set (a polyhedron), and p^* denote its optimal value.

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$(\mathcal{P})$ is a minimization; we seek **valid lower bounds** on $(\mathcal{P})$. *Any ideas?*

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General principle: (i) relax “complicating” constraints; (ii) try “simple” penalty

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For every constraint i , have a **penalty** λ_i

Construct the **lower bound** as the **Lagrangian**:

$$\mathcal{L}(x, \lambda) = c^T x - \sum_{i=1}^m \lambda_i (a_i^T x - b_i) = c^T x - \lambda^T (Ax - b)$$

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Note: we relaxed the complicating constraints, $a_i^T x \text{ (?) } b_i$, and used a linear penalty

Not apriori clear that this will give us very good bounds...

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We want the Lagrangean to give us **a valid lower bound**:

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Deriving Lower Bounds

Summarizing... any $\lambda \in \Lambda$ produces a **valid lower bound**:

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*How can we get a lower bound on the primal's **optimal value** p^* ?*

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Claim

The function $g : \Lambda \rightarrow \mathbb{R}$ defined as:

$$\begin{aligned} g(\lambda) &:= \min_x \mathcal{L}(x, \lambda) \\ \text{s.t. } &x_j \geq 0, \forall j \in J_p \\ &x_j \leq 0, \forall j \in J_n \\ &x_j \text{ free}, \forall j \in J_f \end{aligned} \tag{3}$$

satisfies $g(\lambda) \leq p^$ for any $\lambda \in \Lambda$.*

Note: including the sign constraints on x in this optimization improves the lower bound!

Deriving Lower Bounds

Let us analyze this further:

$$\begin{aligned} g(\lambda) = \min_x \mathcal{L}(x, \lambda) &= \min_x [\lambda^T b + (c^T - \lambda^T A)x] \\ \text{s.t. } x_j &\geq 0, \forall j \in J_p, & \text{s.t. } x_j &\geq 0, \forall j \in J_p, \\ x_j &\leq 0, \forall j \in J_n, & x_j &\leq 0, \forall j \in J_n, \\ x_j &\text{ free}, \forall j \in J_f & x_j &\text{ free}, \forall j \in J_f \end{aligned}$$

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$$g(\lambda) = \begin{cases} \lambda^T b, & \text{if } \lambda^T A_j \leq c_j, \forall j \in J_p \text{ and } \lambda^T A_j \geq c_j, \forall j \in J_n \text{ and } \lambda^T A_j = c_j, \forall j \in J_f \\ -\infty, & \text{otherwise.} \end{cases}$$

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is a **valid lower bound** on the primal **optimal value**: $g(\lambda) \leq p^*$ for any $\lambda \in \Lambda$.

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$$\underset{\lambda \in \Lambda}{\text{maximize}} \ g(\lambda) \tag{4}$$

This is equivalent to the following optimization problem:

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Dual Problem

$$\begin{aligned} & \text{maximize} && \lambda^T b \\ & \text{subject to} && \lambda_i \geq 0, && \forall i \in I_{ge}, \\ & && \lambda_i \leq 0, && \forall i \in I_{le}, \\ & && \lambda_i \text{ free}, && \forall i \in I_{eq}, \\ & && \lambda^T A_j \leq c_j, && \forall j \in J_p, \\ & && \lambda^T A_j \geq c_j, && \forall j \in J_n, \\ & && \lambda^T A_j = c_j, && \forall j \in J_f. \end{aligned} \tag{5}$$

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Definition

This is the **dual** of $(\mathcal{P})$, which we will also refer to as $(\mathcal{D})$. We denote its feasible set with D and its optimal value with d^* .

Note: The dual is also a linear optimization problem!

Primal-Dual Pair

Primal-Dual Pair of Problems

$$\begin{array}{lll} \text{Primal } (\mathcal{P}) & & \\ \text{minimize}_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} & \\ (\lambda_i \rightarrow) & \mathbf{a}_i^T \mathbf{x} \geq b_i, & \forall i \in I_{ge} \\ (\lambda_i \rightarrow) & \mathbf{a}_i^T \mathbf{x} \leq b_i, & \forall i \in I_{le} \\ (\lambda_i \rightarrow) & \mathbf{a}_i^T \mathbf{x} = b_i, & \forall i \in I_{eq} \\ & x_j \geq 0, & \forall j \in J_p \\ & x_j \leq 0, & \forall j \in J_n \\ & x_j \text{ free}, & \forall j \in J_f \\ \text{variables} & \mathbf{x} \in \mathbb{R}^n & \end{array}$$

$$\begin{array}{lll} \text{Dual } (\mathcal{D}) & & \\ \text{maximize}_{\lambda} & \lambda^T \mathbf{b} & \\ & \lambda_i \geq 0, & \forall i \in I_{ge} \\ & \lambda_i \leq 0, & \forall i \in I_{le} \\ & \lambda_i \text{ free}, & \forall i \in I_{eq} \\ & \lambda^T \mathbf{A}_j \leq c_j, & \forall j \in J_p \\ & \lambda^T \mathbf{A}_j \geq c_j, & \forall j \in J_n \\ & \lambda^T \mathbf{A}_j = c_j, & \forall j \in J_f \\ \text{variables} & \lambda \in \mathbb{R}^m. & \end{array}$$

If primal is a minimization, we get the dual as a maximization.

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Primal ($\mathcal{P}$)		Dual ($\mathcal{D}$)	
minimize	$c^T x$	maximize	$\lambda^T b$
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i, \quad \forall i \in I_{ge}$	$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i, \quad \forall i \in I_{le}$	$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i, \quad \forall i \in I_{eq}$	λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0, \quad \forall j \in J_p$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0, \quad \forall j \in J_n$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free, $\forall j \in J_f$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$	variables	$\lambda \in \mathbb{R}^m.$

If primal is a minimization, we get the dual as a maximization.

What if I rewrite the dual problem as a minimization problem, and construct its dual?

*What if instead of constructing a **lower** bound for a **minimization**, I want to get an **upper bound** for a **maximization**?*

Primal-Dual Pair

Primal-Dual Pair of Problems

Primal ($\mathcal{P}$)

minimize $\underset{x}{c^T x}$

$(\lambda_i \rightarrow) \quad a_i^T x \geq b_i, \quad \forall i \in I_{ge}$

$(\lambda_i \rightarrow) \quad a_i^T x \leq b_i, \quad \forall i \in I_{le}$

$(\lambda_i \rightarrow) \quad a_i^T x = b_i, \quad \forall i \in I_{eq}$

$x_j \geq 0, \quad \forall j \in J_p$

$x_j \leq 0, \quad \forall j \in J_n$

$x_j \text{ free}, \quad \forall j \in J_f$

variables $x \in \mathbb{R}^n$

Dual ($\mathcal{D}$)

maximize $\underset{\lambda}{\lambda^T b}$

$\lambda_i \geq 0, \quad \forall i \in I_{ge}$

$\lambda_i \leq 0, \quad \forall i \in I_{le}$

$\lambda_i \text{ free}, \quad \forall i \in I_{eq}$

$\lambda^T A_j \leq c_j, \quad \forall j \in J_p$

$\lambda^T A_j \geq c_j, \quad \forall j \in J_n$

$\lambda^T A_j = c_j, \quad \forall j \in J_f$

variables $\lambda \in \mathbb{R}^m$.

Exercise

Rewrite the dual problem as a minimization problem and construct its dual.

Primal-Dual Pair

Primal-Dual Pair of Problems

Primal ($\mathcal{P}$)

minimize $c^T x$

$$(\lambda_i \rightarrow) \quad a_i^T x \geq b_i, \quad \forall i \in I_{ge}$$

$$(\lambda_i \rightarrow) \quad a_i^T x \leq b_i, \quad \forall i \in I_{le}$$

$$(\lambda_i \rightarrow) \quad a_i^T x = b_i, \quad \forall i \in I_{eq}$$

$$x_j \geq 0, \quad \forall j \in J_p$$

$$x_j \leq 0, \quad \forall j \in J_n$$

$$x_j \text{ free}, \quad \forall j \in J_f$$

variables $x \in \mathbb{R}^n$

Dual ($\mathcal{D}$)

maximize $\lambda^T b$

$$\lambda_i \geq 0, \quad \forall i \in I_{ge}$$

$$\lambda_i \leq 0, \quad \forall i \in I_{le}$$

$$\lambda_i \text{ free}, \quad \forall i \in I_{eq}$$

$$\lambda^T A_j \leq c_j, \quad \forall j \in J_p$$

$$\lambda^T A_j \geq c_j, \quad \forall j \in J_n$$

$$\lambda^T A_j = c_j, \quad \forall j \in J_f$$

variables $\lambda \in \mathbb{R}^m$.

Exercise

Rewrite the dual problem as a minimization problem and construct its dual.

Theorem (For LPs, the dual of the dual is the primal)

If we transform the dual of a linear optimization problem into an equivalent minimization problem and form its dual, we obtain a problem equivalent to the primal.

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize $\underset{x}{c}^T x$

$(\lambda_i \rightarrow) \quad a_i^T x \geq b_i, \quad \forall i \in I_{ge}$

$(\lambda_i \rightarrow) \quad a_i^T x \leq b_i, \quad \forall i \in I_{le}$

$(\lambda_i \rightarrow) \quad a_i^T x = b_i, \quad \forall i \in I_{eq}$

$x_j \geq 0, \quad \forall j \in J_p$

$x_j \leq 0, \quad \forall j \in J_n$

$x_j \text{ free}, \quad \forall j \in J_f$

variables $x \in \mathbb{R}^n$

maximize $\underset{\lambda}{\lambda}^T b$

$\lambda_i \geq 0, \quad \forall i \in I_{ge}$

$\lambda_i \leq 0, \quad \forall i \in I_{le}$

$\lambda_i \text{ free}, \quad \forall i \in I_{eq}$

$(x_j \rightarrow) \quad \lambda^T A_j \leq c_j, \quad \forall j \in J_p$

$(x_j \rightarrow) \quad \lambda^T A_j \geq c_j, \quad \forall j \in J_n$

$(x_j \rightarrow) \quad \lambda^T A_j = c_j, \quad \forall j \in J_f$

variables $\lambda \in \mathbb{R}^m$.

For LPs, dualization in a **very symmetric** construction

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize $c^T x$	maximize $\lambda^T b$
$(\lambda_i \rightarrow) a_i^T x \geq b_i, \quad \forall i \in I_{ge}$	$\lambda_i \geq 0, \quad \forall i \in I_{ge}$
$(\lambda_i \rightarrow) a_i^T x \leq b_i, \quad \forall i \in I_{le}$	$\lambda_i \leq 0, \quad \forall i \in I_{le}$
$(\lambda_i \rightarrow) a_i^T x = b_i, \quad \forall i \in I_{eq}$	$\lambda_i \text{ free}, \quad \forall i \in I_{eq}$
$x_j \geq 0, \quad \forall j \in J_p$	$(x_j \rightarrow) \lambda^T A_j \leq c_j, \quad \forall j \in J_p$
$x_j \leq 0, \quad \forall j \in J_n$	$(x_j \rightarrow) \lambda^T A_j \geq c_j, \quad \forall j \in J_n$
$x_j \text{ free}, \quad \forall j \in J_f$	$(x_j \rightarrow) \lambda^T A_j = c_j, \quad \forall j \in J_f$
variables $x \in \mathbb{R}^n$	variables $\lambda \in \mathbb{R}^m$

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$\overset{x}{c}^T x$		maximize	$\overset{\lambda}{\lambda}^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$		variables	$\lambda \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$\mathbf{c}^T \mathbf{x}$		maximize	$\boldsymbol{\lambda}^T \mathbf{b}$	
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j = c_j,$	$\forall j \in J_f$
variables	$\mathbf{x} \in \mathbb{R}^n$		variables	$\boldsymbol{\lambda} \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize $\overset{x}{c^T x}$		maximize $\overset{\lambda}{\lambda^T b}$	
$(\lambda_i \rightarrow) \quad a_i^T x \geq b_i, \quad \forall i \in I_{ge}$		$\lambda_i \geq 0, \quad \forall i \in I_{ge}$	
$(\lambda_i \rightarrow) \quad a_i^T x \leq b_i, \quad \forall i \in I_{le}$		$\lambda_i \leq 0, \quad \forall i \in I_{le}$	
$(\lambda_i \rightarrow) \quad a_i^T x = b_i, \quad \forall i \in I_{eq}$		$\lambda_i \text{ free}, \quad \forall i \in I_{eq}$	
$x_j \geq 0, \quad \forall j \in J_p$		$(x_j \rightarrow) \quad \lambda^T A_j \leq c_j, \quad \forall j \in J_p$	
$x_j \leq 0, \quad \forall j \in J_n$		$(x_j \rightarrow) \quad \lambda^T A_j \geq c_j, \quad \forall j \in J_n$	
$x_j \text{ free}, \quad \forall j \in J_f$		$(x_j \rightarrow) \quad \lambda^T A_j = c_j, \quad \forall j \in J_f$	
variables $x \in \mathbb{R}^n$		variables $\lambda \in \mathbb{R}^m$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization
- Dual objective: inner product of **primal r.h.s.** and **dual decisions**

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$\underset{x}{c^T x}$		maximize	$\underset{\lambda}{\lambda^T b}$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$		variables	$\lambda \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization
- Dual objective: inner product of **primal r.h.s.** and **dual decisions**
- Depending on **primal constraint type** ($\geq, \leq, =$), add constraint on **sign of dual decision**

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$c^T x$		maximize	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$		variables	$\lambda \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization
- Dual objective: inner product of **primal r.h.s.** and **dual decisions**
- Depending on **primal constraint type** ($\geq, \leq, =$), add constraint on **sign of dual decision**
- Depending on **sign of primal decision**, set up **constraint on dual decisions**

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{lcl}
 \text{minimize}_{\mathbf{x}} & / & \text{maximize}_{\mathbf{x}} \quad \mathbf{c}^T \mathbf{x} \\
 & & (\lambda \rightarrow) \quad \mathbf{Ax} \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \end{array} \mathbf{b} \\
 & & \mathbf{x} \begin{array}{l} \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} \mathbf{0}
 \end{array} \tag{7}$$

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{lcl}
 \text{minimize}_{\mathbf{x}} & / & \text{maximize}_{\mathbf{x}} \quad c^T \mathbf{x} \\
 (\lambda \rightarrow) & & A \mathbf{x} \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} b \\
 & & \mathbf{x} \begin{array}{l} \geq \\ \leq \\ \geq \\ \leq \\ \geq \\ \leq \end{array} 0
 \end{array} \tag{7}$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{lcl} \underset{x}{\text{minimize}} & / & \underset{x}{\text{maximize}} \quad c^T x \\ & & (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ & & \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i $\textcircled{?}$ 0.

To determine the signs $\textcircled{?}$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

– primal minimization, “ $\leq$ ” constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs (?) , ask yourself this Q:

*What happens to the **optimal objective value** if I **increase** the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/ **maximization**? Is constraint $\leq$ or $\geq$?

– primal minimization, “ $\leq$ ” constraint: objective would decrease

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs (?) , ask yourself this Q:

*What happens to the **optimal objective value** if I **increase** the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs (?) , ask yourself this Q:

*What happens to the **optimal objective value** if I **increase** the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/ **maximization**? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs $(?)$, ask yourself this Q:

*What happens to the **optimal objective value** if I **increase** the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/ **maximization**? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs $(?)$, ask yourself this Q:

*What happens to the **optimal objective value** if I **increase** the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/ **maximization**? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{ll} \underset{x}{\text{minimize}} & / \quad \underset{x}{\text{maximize}} \quad c^T x \\ & (\lambda \rightarrow) \quad Ax \preceq b \\ & \quad \quad \quad x \preceq 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i (?) 0.

To determine the signs (?), ask yourself this Q:

What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?

Powerful rule of thumb, which we will prove later:

λ_j is the slope of the optimal objective value w.r.t. the right-hand-side b_j

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$

Note the relationship between: (i) the primal problem type, (ii) the constraint type, and (iii) the sign of the dual variable: **given any two, you can always figure out the third!**

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \end{array} 0 \end{array} \quad (8)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$

R3: In the dual, add a **constraint** for every primal variable x_j
If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j \text{ (?) } c_j$.

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \geq \\ \leq \\ \leq \end{array} 0 \end{array} \quad (8)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i $(?)$ 0.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
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- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
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R3: In the dual, add a **constraint** for every primal variable x_j
If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j$ $(?)$ c_j .

Determine $(?)$ using rules in [R2]: treat dual as primal, c_j as r.h.s., x_j as dual variable

- dual maximization, $x_j \geq 0$:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \end{array} 0 \end{array} \quad (8)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i (?) 0.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$

R3: In the dual, add a **constraint** for every primal variable x_j
If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j$ (?) c_j .

Determine (?) using rules in [R2]: treat dual as primal, c_j as r.h.s., x_j as dual variable

- dual maximization, $x_j \geq 0$: $\lambda^T A_j \leq c_j$
- dual maximization, $x_j \leq 0$:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{c} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{c} \leq \\ \geq \\ \leq \\ \geq \end{array} b \\ \quad \quad \quad x \begin{array}{c} \geq \\ \leq \end{array} 0 \end{array} \quad (8)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$

R3: In the dual, add a **constraint** for every primal variable x_j
If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j \text{ (?) } c_j$.

Determine (?) using rules in [R2]: treat dual as primal, c_j as r.h.s., x_j as dual variable

- dual maximization, $x_j \geq 0$: $\lambda^T A_j \leq c_j$
- dual maximization, $x_j \leq 0$: $\lambda^T A_j \geq c_j$
- dual minimization, $x_j \geq 0$:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \geq \\ \geq \\ \geq \\ \geq \\ \geq \end{array} 0 \end{array} \quad (8)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i $(?)$ 0.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$

R3: In the dual, add a **constraint** for every primal variable x_j
If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j$ $(?)$ c_j .

Determine $(?)$ using rules in [R2]: treat dual as primal, c_j as r.h.s., x_j as dual variable

- dual maximization, $x_j \geq 0$: $\lambda^T A_j \leq c_j$
- dual maximization, $x_j \leq 0$: $\lambda^T A_j \geq c_j$
- dual minimization, $x_j \geq 0$: $\lambda^T A_j \geq c_j$
- dual minimization, $x_j \leq 0$:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} 0 \end{array} \quad (8)$$

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- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
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If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j \text{ (?) } c_j$.

Determine (?) using rules in [R2]: treat dual as primal, c_j as r.h.s., x_j as dual variable

- dual maximization, $x_j \geq 0$: $\lambda^T A_j \leq c_j$
- dual maximization, $x_j \leq 0$: $\lambda^T A_j \geq c_j$
- dual minimization, $x_j \geq 0$: $\lambda^T A_j \geq c_j$
- dual minimization, $x_j \leq 0$: $\lambda^T A_j \leq c_j$

Mnemonic Rules for Constructing **Any** Dual

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i $(?)$ 0.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization: " $\leq$ " constraint $\rightarrow \lambda_i \leq 0$. " $\geq$ " constraint $\rightarrow \lambda_i \geq 0$
- primal maximization: " $\leq$ " constraint $\rightarrow \lambda_i \geq 0$. " $\geq$ " constraint $\rightarrow \lambda_i \leq 0$

R3: In the dual, add a **constraint** for every primal variable x_j

If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j$ $(?)$ c_j .

Determine $(?)$ using rules in [R2]: treat dual as primal and " x_j " as dual variable

- dual maximization: $x_j \geq 0 \Rightarrow \lambda^T A_j \leq c_j$. $x_j \leq 0 \Rightarrow \lambda^T A_j \geq c_j$
- dual minimization: $x_j \geq 0 \Rightarrow \lambda^T A_j \geq c_j$. $x_j \leq 0 \Rightarrow \lambda^T A_j \leq c_j$

$$(\mathcal{P}) \quad \min_{x_1, x_2} \quad c_1^T x_1 + c_2^T x_2$$

$$\text{s.t.} \quad A_{11} x_1 + A_{12} x_2 \geq b_1,$$

$$A_{21} x_1 + A_{22} x_2 = b_2,$$

$$x_1 \geq 0,$$

$$x_2 \text{ free.}$$

Mnemonic Rules for Constructing **Any** Dual

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

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- primal minimization: " $\leq$ " constraint $\rightarrow \lambda_i \leq 0$. " $\geq$ " constraint $\rightarrow \lambda_i \geq 0$
- primal maximization: " $\leq$ " constraint $\rightarrow \lambda_i \geq 0$. " $\geq$ " constraint $\rightarrow \lambda_i \leq 0$

R3: In the dual, add a **constraint** for every primal variable x_j

If x_j is **free**, write this as $\lambda^T A_j = c_j$. Otherwise: $\lambda^T A_j$ $(?)$ c_j .

Determine $(?)$ using rules in [R2]: treat dual as primal and " x_j " as dual variable

- dual maximization: $x_j \geq 0 \Rightarrow \lambda^T A_j \leq c_j$. $x_j \leq 0 \Rightarrow \lambda^T A_j \geq c_j$
- dual minimization: $x_j \geq 0 \Rightarrow \lambda^T A_j \geq c_j$. $x_j \leq 0 \Rightarrow \lambda^T A_j \leq c_j$

$$(\mathcal{P}) \quad \min_{x_1, x_2} \quad c_1^T x_1 + c_2^T x_2$$

$$\text{s.t.} \quad A_{11} x_1 + A_{12} x_2 \geq b_1,$$

$$A_{21} x_1 + A_{22} x_2 = b_2,$$

$$x_1 \geq 0,$$

$$x_2 \text{ free.}$$

$$(\mathcal{D}) \quad \max_{\lambda_1, \lambda_2} \quad b_1^T \lambda_1 + b_2^T \lambda_2$$

$$\text{s.t.} \quad A_{11}^T \lambda_1 + A_{21}^T \lambda_2 \leq c_1,$$

$$A_{12}^T \lambda_1 + A_{22}^T \lambda_2 = c_2,$$

$$\lambda_1 \geq 0,$$

$$\lambda_2 \text{ free.}$$

Some Quick Results

Theorem ("Duals of equivalent primals")

If we transform a primal P_1 into an equivalent formulation P_2 by:

- *replacing a free variable x_i with $x_i = x_i^+ - x_i^-$,*
- *replacing an inequality with an equality by introducing a slack variable,*
- *removing linearly dependent rows a_i^T for a **feasible** LP in standard form,*

*then the duals of (P_1) and (P_2) are **equivalent**, i.e., they are either both infeasible or they have the same optimal objective.*

Weak duality

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Weak duality

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Theorem (Weak duality)

If x is feasible for ($\mathcal{P}$) and λ is feasible for ($\mathcal{D}$), then $\lambda^T b \leq c^T x$.

Proof. Trivially true from our construction – omitted.

Implications of Weak Duality

Corollary

The following results hold:

- (a) If the optimal objective in $(\mathcal{P})$ is $-\infty$, then $(\mathcal{D})$...*
- (b) If the optimal objective in $(\mathcal{D})$ is $+\infty$, then $(\mathcal{P})$...*

Implications of Weak Duality

Corollary

The following results hold:

- (a) If the optimal objective in $(\mathcal{P})$ is $-\infty$, then $(\mathcal{D})$... must be infeasible.*
- (b) If the optimal objective in $(\mathcal{D})$ is $+\infty$, then $(\mathcal{P})$...*

Implications of Weak Duality

Corollary

The following results hold:

- (a) If the optimal objective in $(\mathcal{P})$ is $-\infty$, then $(\mathcal{D})$... must be infeasible.*
- (b) If the optimal objective in $(\mathcal{D})$ is $+\infty$, then $(\mathcal{P})$... must be infeasible.*
- (c) If $x \in P$ and $\lambda \in D$, then:*

$$c^T x - p^* \leq c^T x - \lambda^T b \quad \textbf{and} \quad d^* - \lambda^T b \leq c^T x - \lambda^T b.$$

Implications of Weak Duality

Corollary

The following results hold:

(a) If the optimal objective in $(\mathcal{P})$ is $-\infty$, then $(\mathcal{D})$... must be infeasible.

(b) If the optimal objective in $(\mathcal{D})$ is $+\infty$, then $(\mathcal{P})$... must be infeasible.

(c) If $x \in P$ and $\lambda \in D$, then:

$$c^T x - p^* \leq c^T x - \lambda^T b \quad \textbf{and} \quad d^* - \lambda^T b \leq c^T x - \lambda^T b.$$

*(d) If $x \in P$, $\lambda \in D$, and $\lambda^T b = c^T x$, then x **optimal** for $(\mathcal{P})$ and λ **optimal** for $(\mathcal{D})$.*

Implications of Weak Duality

Corollary

The following results hold:

(a) If the optimal objective in $(\mathcal{P})$ is $-\infty$, then $(\mathcal{D})$... must be infeasible.

(b) If the optimal objective in $(\mathcal{D})$ is $+\infty$, then $(\mathcal{P})$... must be infeasible.

(c) If $x \in P$ and $\lambda \in D$, then:

$$c^T x - p^* \leq c^T x - \lambda^T b \quad \textbf{and} \quad d^* - \lambda^T b \leq c^T x - \lambda^T b.$$

*(d) If $x \in P$, $\lambda \in D$, and $\lambda^T b = c^T x$, then x **optimal** for $(\mathcal{P})$ and λ **optimal** for $(\mathcal{D})$.*

(c) and (d) provide (sub)optimality certificates, but...

How do we know that the gaps in (c) are not very large?

How do we know that x and λ satisfying (d) even exist?